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Deep neural network-based pulse shape discrimination of neutrons and γ-rays in organic scintillation detectors

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Abstract. Organic scintillation detectors like BC501A, BC519, NE213, etc. have an inherent ability to classify neutrons and γ-rays through a process known as pulse shape discrimination (PSD). We developed a deep neural network (DNN)-based machine learning algorithm to discriminate neutrons/γ-rays. The algorithm was trained with data obtained from a BC501A detector considering the Cf-252 source. Further, to assess the performance of the DNN-based PSD algorithm, the algorithm was tested with an independent data set acquired with a different source-detector set-up, namely, BC501 detector with Am–Be source and a different digitiser. Results indicate that our proposed algorithm can successfully discriminate the neutrons and γ-rays with reasonably good accuracy for the independent data set.

Keywords. Nuclear security and safety; scintillation detectors; neutrons; γ-rays; pulse shape discrimination; deep neural network.

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1. Introduction

Accurate pulse shape discrimination (PSD) is essential for any detector that is sensitive to multiple radiations where a minimal false alarm is of paramount importance such as for applications in nuclear non-proliferation, nuclear security and nuclear safeguards. It is critical to have fast neutron detectors like BC501, EJ399, etc., for neutron security and safeguards applications as these detectors are both sensitive to fast neutrons and γ-rays and thermal neutron detectors, like GGAG, CLYC, LiI, Li-Glass, etc., where it is often desirable to distinguish various types of radiation on a waveform by waveform basis. Poor PSD demands long measurement times which eventually leads to an inappropriate characterisation of radiations leading to false identification of the radioactive source. For applications like nuclear non-proliferation, safeguards and nuclear medicine, robust and fast PSD systems are of the highest priority as false identification and false alarm can be detrimental. Several well-known and time-tested PSD methods are available [1,2] and it has been established that digital systems are advantageous over analog PSD systems. The widely popular digital PSD method, charge integration (CI) [3–7] which is proved to be reasonably accurate for neutrons of high energies, exhibits poor performance at low energies. As the PSD is a classification problem, it becomes a common problem for machine learning algorithms to make it a computationally more efficient and robust method.

A few but significantly important studies are reported on PSD using machine learning. A study reported several standard PSD techniques applied to the BC501 scintillation detector and compared them to pattern recognition sigmoidal neural network (NN) [8]. Results revealed that the NN method of n-γ discrimination had a lead compared to the two-dimensional standard PSD methods (charge ratio and charge rise time), especially in regions of small energy deposition. Another study reported PSD using support vector machine (SVM)

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algorithm for an EJ309 detector [9]. It showed that the SVM method of PSD leads to improved detection performance compared to the traditional CI method. An artificial neural network (ANN) was also employed to obtain n-γ-ray discrimination for high purity germanium detector (HPGe) detector using the time-of-flight (ToF) method [10]. The study shows good agreement with ToF in discriminating γ-rays and neutrons using the HPGe detector. The testing was performed only for neutrons and γ-rays separately and not in a mixed radiation environment. Fu *et al* [11] developed two NN-based algorithms, namely fully connected neural network (FC-NN) and recurrent neural network (RNN) for PSD of an organic stilbene scintillation detector. Both the NN-based approaches successfully separated single pulses from piled-up pulses in comparison to the CI approach regardless of the pulse amplitude and achieved a low fraction of misclassified piled-up pulses. In another study, five different PSD procedures for neutron/γ-ray discrimination were tested, namely, tail-to-total (TtoT) area, Gatti DeMartini, principal component analysis (PCA), semi-supervised (SS) and unsupervised (US) [12]. All the classifiers perform optimally at higher energy levels but the true positive rate diminishes rapidly at lower energy levels. A recent study presented two machine learning techniques applied to n-γ-ray discrimination for ^{7}Li-enriched CLYC detector [7]. The two methods used were supervised artificial neural network (ANN) and unsupervised (k-mean++) clustering algorithms to classify neutrons and γ-rays and compared the algorithms' discrimination capability with the traditional CI method. It showed that both ANN and k-mean++ methods were capable of providing good n-γ-ray discrimination in the energy range of interest. The ANN for the supervised learning provides superior n-γ-ray discrimination compared to the CI method. The k-mean++ cluster algorithm is capable of separating neutrons and γ-rays in CLYC detectors albeit with more false identification. A recent study introduces a novel application of the KNN algorithm for n-γ-ray discrimination for stilbene and EJ299 [13] detectors. The KNN algorithm performs regression to identify the pulse shape parameters as opposed to the classification-based approach. Results indicate that the KNN algorithm (regressor model) provides tighter distribution in pulse features which leads to a higher figure of merit (FOM) for even low light output ranges. It is worth noting that in spite of these studies exhibiting promising results, no independent data sets were tested (data collected with a different source-detector set-up) for the performance of any of the proposed algorithms.

The study presented in this paper focusses on finding the PSD capability of a deep neural network (DNN) algorithm in discriminating n-γ for two neutron detectors (BC501A and BC501) in a mixed radiation environment and comparing them to the traditional PSD method. This algorithm was trained using classified data obtained from a BC501A liquid scintillator [14] considering Cf-252 source and a digitiser. The experiment is performed at IIT Roorkee, India. Further, to check the accuracy and discrimination capability of the proposed algorithm, we tested this algorithm on an independent data set obtained from BC501 liquid scintillator [14] considering Am–Be source with a different digitiser. The experiment is performed at BARC, Mumbai, India.

2. Methods

2.1 *Experimentation for training data*

The investigation utilises data acquired from a simple testing experiment for developing a machine learning algorithm that can discriminate neutrons and γ-rays. A 1.3 μCi Cf-252 source was placed at a distance of 12 cm from a $3'' \times 3''$ BC501A detector. The input voltage applied to the PMT was -1400 V. The waveforms of neutron and γ-ray obtained from the digitiser are shown in figure 1. Classified pulses were acquired for 45 min using a 1 GHz, DT5751 digitiser. The digitiser calculated the PSD based on the charge integration method. The charges under the short gate and the long gate are integrated and used in the following formula to calculate the PSD as shown in eq. (1) where Q_s is the charge integrated under the short gate and Q_l is the charge integrated under the long gate.

$$\mathrm{PSD} = 1 - Q_s/Q_l. \tag{1}$$

The scatter plot of neutrons and γ-rays of BC501A detector obtained using Cf-252 source is drawn between the PSD parameter described above vs. the channel numbers of the ADC as shown in figure 2. To get classified pulses, first raw data with no classification were obtained using the digitiser and then the scatter plot was generated. Appropriate PSD values were chosen to isolate the gamma band and the neutron band based on the position of bands of γ-rays and neutrons from the scatter plot. Then PSD cuts were applied before starting a fresh acquisition, and filtered data sets for γ-rays and neutrons were saved subsequently. The total and tail area of the γ-ray and neutron pulses thus obtained are used for training DNN algorithm. Please note that this experiment, labelled as training data (TD), was conducted at Radiation Detectors and Spectroscopy Laboratory, Indian Institute of Technology Roorkee, India.

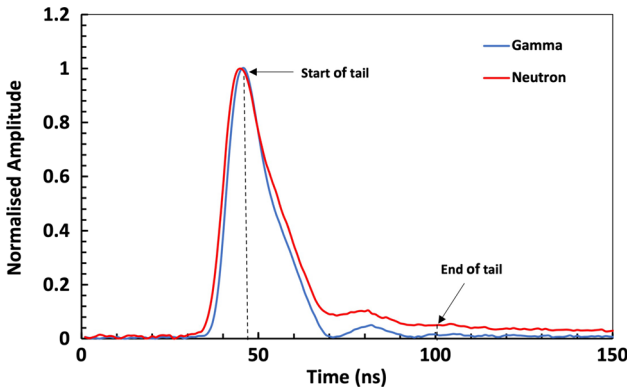

Figure 1. Normalised neutron and γ-ray pulses obtained from the digitizer.

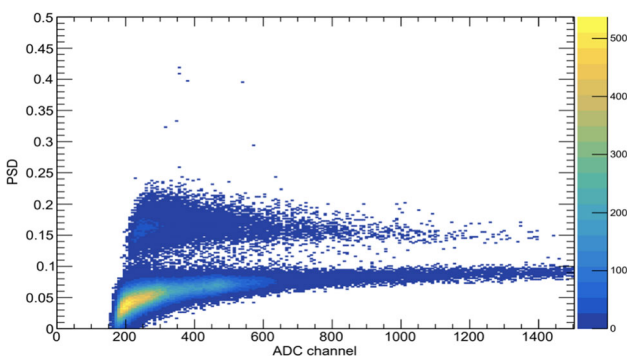

Figure 2. Scatter plot of BC501A drawn between PSD and ADC channel. The upper band corresponds to events corresponding to neutrons and the lower band represents events due to γ-rays. The tail area and total area along with the classification of these pulses are used as training input for our proposed DNN-based PSD algorithm.

2.2 *Experimentation for independent data set*

To prepare the independent data set, a different experiment is performed. An Am–Be source of activity 30 mCi was placed 6″ away from the BC501 detector of 10 mm thickness and 2″ diameter. The input voltage applied to the PMT was 1000 V. Classified pulses are acquired for 10 min through a 0.25 GHz, DT 5790 digitiser. This experiment is labelled as independent data (ID) and was conducted at Technical Physics Laboratory, BARC, Mumbai, India. Table 1 summarises the technical specifications of both the training and independent data sets used in the present study.

2.3 *Deep neural network (DNN) pulse shape discriminator*

Machine learning algorithms are powerful tools for various classification problems [7–9,15,16]. Supervised learning techniques, such as artificial neural networks (ANNs), have been used for neutron/γ discrimination in liquid scintillators and have an advantage over other standard PSD techniques for liquid scintillators, such as CI method [3–7], time-of-flight method [17] and gradient descent method [18]. Since our proposed DNN algorithm is a supervised model, training on a base model is necessary. In this study, we chose the CI method as the base model for training our DNN algorithm. We have used the tail and total area of the pulses obtained from our experiment, labelled as TD, as the training input to the DNNs, and their classifications of the pulses (either neutron/γ) as the output. Our data set comprises 46248 such input–output pairs with 33248 neutrons and 13000 γ-rays classified pulses.

A dense, fully-connected, feed-forward DNN is chosen for this application. An example of the DNN of this type is as follows:

$$h_1 = \varphi_1\,(W_1\mathcal{A}_{\text{in}} + b_1) \tag{2}$$

$$h_2 = \varphi_2\,(W_2 h_1 + b_2) \tag{3}$$

$$h_3 = \varphi_3\,(W_3 h_2 + b_3) \tag{4}$$

$$C_{\text{out}} = (W_4 h_3 + b_4)\,. \tag{5}$$

Here $\mathcal{A}_{\text{in}}(1\times 2)$ denotes the input vector, $C_{\text{out}}(1\times 1)$ is the predicted output vector, $W_1(2\times 64)$, $W_2(64\times 32)$, $W_3(32\times 16)$, $W_4(16\times 1)$ are matrices of trainable weights, $b_1(1\times 64)$, $b_2(1\times 32)$, $b_3(1\times 16)$, $b_4(1\times 1)$ are 'bias vectors', $\varphi_1, \varphi_2, \varphi_3$ are the non-linear activation functions and $h_1(1\times 64)$, $h_2(1\times 32)$ and $h_3(1\times 16)$ are the 'hidden' vectors whose scalar components are called 'neurons'. Note that activation functions are only applied to the hidden vectors. For the present investigation, a neural network package called Keras [19] has been used, which is a high-level wrapper around TensorFlow [20] written in python. The neural network proposed here was chosen to achieve optimal precision and accuracy for neutron/γ-rays PSD. To achieve 99.9% accuracy, we tried several combinations of layers and neurons by varying the number of epochs and activation functions.

Table 1. Technical specifications of datasets.

Data Set	Detector	Source	Frequency of digitiser	Number of pulses	Experiments conducted at
Training data (TD)	BC501A″Φ, 3″l)	Cf-252	1 GHz	46248	IIT Roorkee
Independent data (ID)	BC501 (2″Φ, 10 mm l)	Am-Be	0.25 GHz	9031	BARC, Mumbai

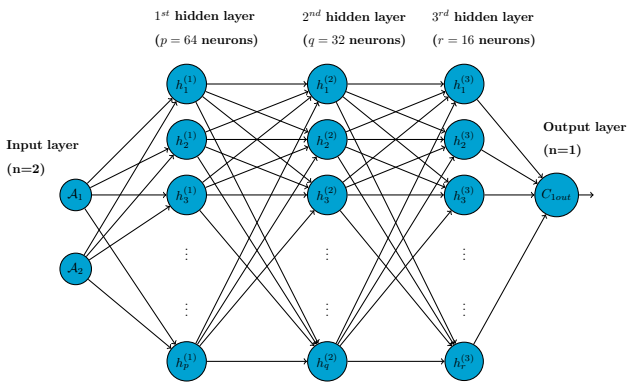

Figure 3. Schematic of deep neural network architecture.

The tail and total area per waveform sample are the two inputs to the DNN. The tail area is considered from the peak amplitude of the waveform till the wave falls up to 100 ns in the x-axis as shown in figure 1. We have used Relu [21] as the activation function in all three hidden layers. At the output, we use a sigmoid activation function [22] to classify the neutrons and the γ-rays based on the tail and total area. Two hundred training epochs took approximately 5 min on a dual-core $i5$ CPU workstation using an adaptive learning rate optimiser adam [23] and binary cross-entropy [24] as the loss function. The schematic diagram of the DNN algorithm is shown in figure 3.

The following terminology is used to describe different data sets:

1. Training data set: A random 70% of the data set TD is used to train the DNN over a series of epochs.
2. Testing data set: A random 30% of the data set TD is used to probe the accuracy of the DNN after they have been trained. The DNN is never trained on this 30% data set during any epoch.
3. Validation data set: During each epoch, a random 10% of the training data set TD remains unused during a single epoch and is used to track DNN generalisation as epochs progress during training.

The batch size of the training data for the DNN model is 32374 samples, validation samples consisted of 3237 samples and testing consisted of 13874 samples. The dataset was normalised using the 'max' scalar. The neutrons and the gammas were allotted a value of 1 and 0 respectively. The hyper-parameters for the DNN are summarised in table 2. The training and validation losses for the DNN algorithm are shown in figure 4.

3. Results and discussion

To evaluate the robustness and generalisation of our DNN algorithm, we use the trained algorithm to discriminate the neutrons and γ-rays of an independent data set (ID) discussed in §2.2. Notice that the independent data set was never used for training the DNN and are, therefore, a good measure of the generality of the DNN algorithm for the discrimination of the neutrons and γ-rays for fast neutron detection purposes. In this study, we quantify the PSD capability and evaluate the classification accuracy of the neutrons and γ-rays by our DNN model with a confusion matrix. The confusion matrix gives the exact number of neutrons and γ-rays predicted by our proposed DNN model and also shows a clear comparison of performance with a base model (CI method in the present case). Confusion matrices are plotted for the testing data set (30% of the training data) and independent data set ID. A confusion matrix is a table that is used to define the performance of a classification algorithm. It consists of four basic characteristics (numbers) that are used to define the measurement metrics of the classifier. These four numbers are:

1. TG (true positive): TG represents the number of correctly classified γ-rays.
2. TN (true negative): TN represents the number of correctly classified neutrons.
3. FG (false positive): FG represents the number of misclassified γ-rays, but they actually are neutrons. FG is also known as a Type-I error.
4. FN (false negative): FN represents the number of misclassified neutrons, but they actually are γ-rays. FN is also known as a Type-II error.

Performance metrics of an algorithm are accuracy, precision, recall and F1 score, which are calculated on the basis of the above-stated TG, TN, FG and FN. The accuracy of an algorithm is the ratio of correctly classified γ-rays and neutrons (TG+TN) to the total number of

Table 2. Hyperparameters of ML models.

Hyperparameters	DNN
Training samples	32374
Validation samples	3237
Testing samples	13874
Epoch	200
Hidden layers	3
Neurons in first hidden layer	64
Neurons in second hidden layer	32
Neurons in third hidden layer	16
Activation functions for first hidden layer	Relu
Activation functions for second hidden layer	Relu
Activation functions for third hidden layer	Relu
Output activation function	Sigmoid
Optimiser	Adam
Loss function	Binary-cross-entropy

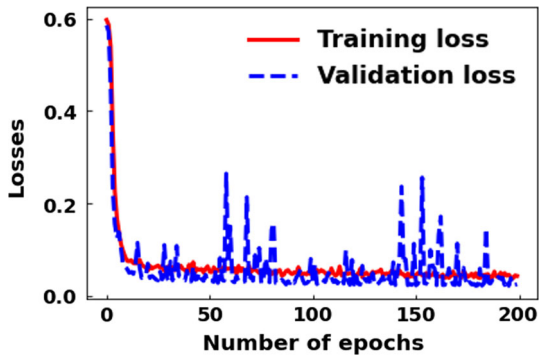

Figure 4. Comparison of the training and validation loss of DNN.

γ-rays and neutrons (TG+TN+FG+FN).

$$\text{Accuracy} = \frac{\text{TG} + \text{TN}}{\text{TG} + \text{TN} + \text{FG} + \text{FN}}. \tag{6}$$

The precision of neutrons (P_N) is the ratio of the correctly classified neutrons (TN) to the total number of predicted neutrons (TN+FN). Similarly, the precision of γ-rays (P_G) is the ratio of the correctly classified γ (TG) to the total number of predicted γ (TG+FG).

$$P_N = \frac{\text{TN}}{\text{TN} + \text{FN}}, \tag{7}$$

$$P_G = \frac{\text{TG}}{\text{TG} + \text{FG}}. \tag{8}$$

The recall metric for neutrons (R_N) is the ratio of the accurately classified neutrons to the total number of actual neutrons. Similarly, recall for γ-rays (R_G) is defined as the ratio of the correctly classified γ-rays (TG) to the total number of actual γ-rays.

$$R_G = \frac{\text{TG}}{\text{TG} + \text{FN}}, \tag{9}$$

$$R_N = \frac{\text{TN}}{\text{TN} + \text{FG}}. \tag{10}$$

Precision can be interpreted as a measure of quality whereas recall represents a measurement of quantity. For any algorithm, a higher precision signifies the capability of the algorithm to return more relevant results than irrelevant ones, whereas high recall means that the algorithm returns most of the relevant results (whether or not irrelevant ones are also returned).

The F1 score states the equilibrium between precision and recall by computing the harmonic mean of precision and recall. We define the F1 score for neutrons ($F1_N$) and γ-rays ($F1_G$) as follows:

$$F1_N = \frac{2 \times P_N \times R_N}{P_G + R_N}, \tag{11}$$

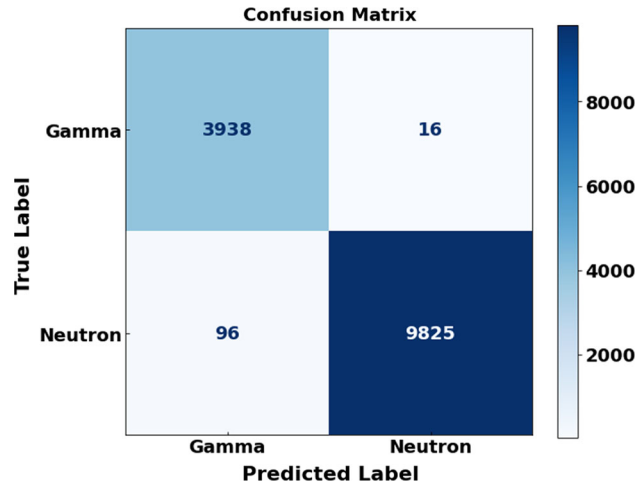

Figure 5. Confusion matrix for the testing data set.

$$F1_G = \frac{2 \times P_G \times R_G}{P_G + R_G}. \tag{12}$$

The 'True Label' in the following confusion matrices is the classification of neutrons and γ-rays of the data sets under consideration using the CI method and the 'Predicted Label' represents the classification of the same using the DNN-based PSD algorithm. For the testing data set as demonstrated in figure 5, the true neutron count is $96 + 9825 = 9921$, whereas the true γ count is $3938 + 16 = 3954$. The accuracy of our DNN model in classifying neutron and γ-rays on this testing data set is 99.1%. The precision for neutrons and γ-rays are 100% and 98% respectively. Correspondingly, the recall for neutrons and γ-rays are 99% and 100%. The $F1_N$ is 99% and $F1_G$ is 99%. All the metrics indicate that our DNN model predicts neutrons and γ-rays for the testing data set accurately with negligible (96) false prediction for neutrons and 16 false prediction for γ-rays. The classification capability of our DNN-based algorithm for the independent data set is qualitatively represented by the scatter plot of ID in figure 6.

Since the tail area and total area of the pulses are taken as the input for our proposed DNN-based PSD algorithm, the scatter plot shows the neutrons and γ-rays based on the tail area and total area. The upper cluster corresponds to events due to neutrons and the lower cluster represents events due to γ-rays. It can be seen that our proposed algorithm can very well classify neutrons and γ-rays for the ID. The scatter plot generated by the DNN-based PSD method (figure 6) for the ID shows a better separation of bands than that generated by the CI method (figure 2) for the TD, i.e. training data set. One of the reasons for this can be attributed to the better discrimination capability of the DNN-based method over the CI method. Another reason could be the different nature of the two graphs. As the axes of

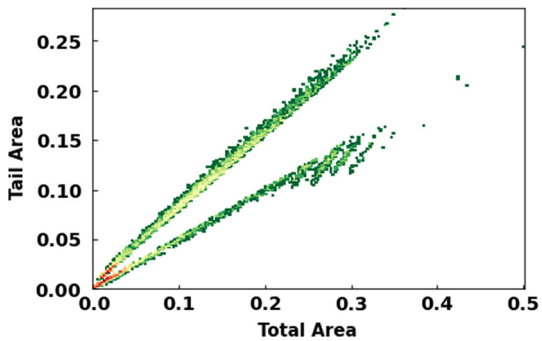

Figure 6. Scatter plot of the tail area vs. total area of the digitiser pulses for the ID data set using DNN-PSD method.

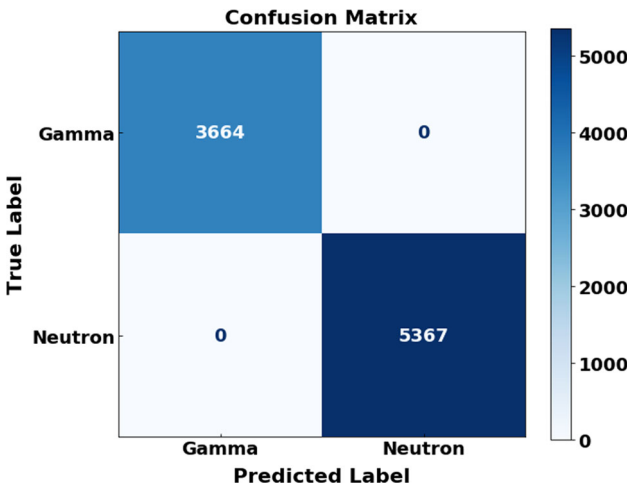

Figure 7. Confusion matrix for the ID data set.

both graphs are not the same, we cannot compare these two graphs directly. Also, the number of pulses used in generating the scatter plot by the DNN-based method is lower than that used in the CI method.

The quantification of the performance of our DNN-based PSD algorithm to discriminate neutrons and γ-rays for ID is demonstrated in the form of a confusion matrix in figure 7. We can see that the true and the predicted counts of neutrons are 5367 and that of γ-rays are 3664. Hence, the precision of the DNN-based PSD model for neutron and γ-rays is 100% in the case of ID. The recall for neutrons and γ-rays is also 100%. The accuracy and the F1-scores are 100% which means that our DNN model can make excellent classification between neutrons and γ-rays for the ID.

4. Conclusion

The proposed DNN algorithm demonstrates good discrimination capability for fast neutron detectors. Results indicate that through proper training, DNN can successfully discriminate neutrons and γ-rays. Additionally, our proposed DNN-PSD algorithm was able to predict neutrons and γ-rays with good accuracy for an independent data set with different source-detector and digitiser set-up. We have presented the classification capability of our algorithm using confusion matrices with different parameters to assess the performance of our DNN-based algorithm.

The results are promising enough to motivate futuristic research in developing deep learning-based algorithms that can be applied to discriminate multiple radiations in mixed radiation fields and other detector set-ups for varied energy ranges, making it a good and fast post-processing application in various fields like nuclear security, nuclear medicine and space radiation exploration.

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